Zepto Inventory Analysis: Unlocking Business Insights from Data

A deep dive into inventory patterns, pricing strategies, and customer value across India's fastest 10-minute grocery delivery platform.

FAST COMMERCE

Why Zepto's Inventory Data Matters

Zepto has revolutionised grocery delivery in India with its promise of 10-minute delivery times. Understanding inventory patterns is crucial for maintaining this lightning-fast service whilst protecting margins and customer satisfaction.

This analysis examines over 3,000 SKUs from a Kaggle dataset, revealing critical insights about pricing strategy, stock management, and value creation.

Analysis Goals

- Optimise pricing and discount strategies
- Identify stock replenishment opportunities
- Maximise customer value perception
- Balance revenue growth with cost control

3K+

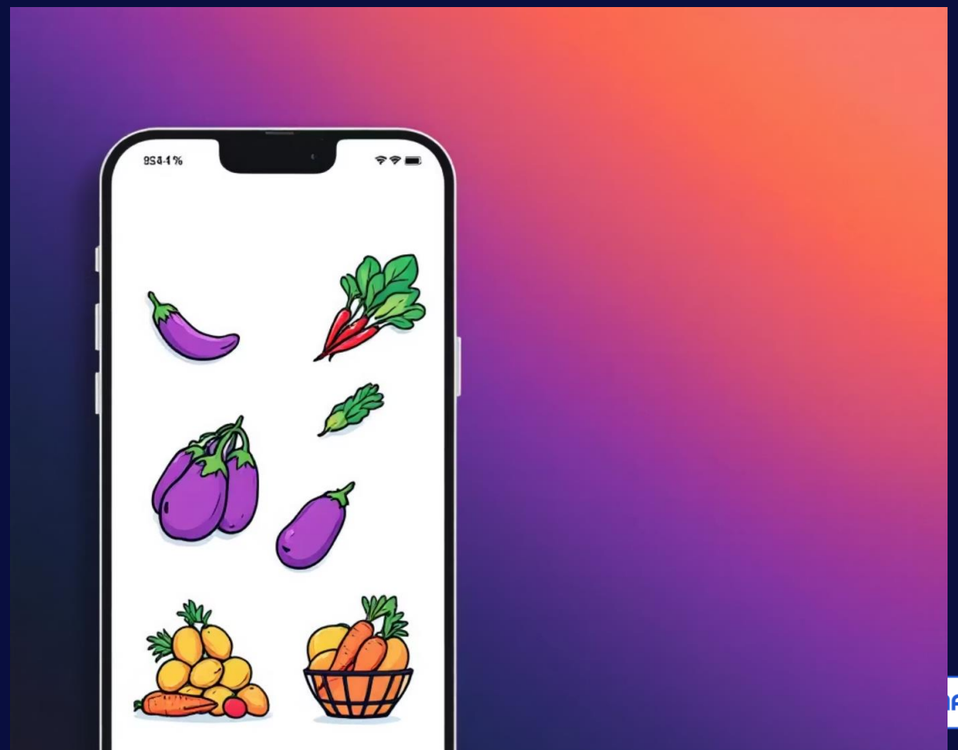
Product SKUs

Analysed across categories

10min

Delivery Promise

Ultra-fast logistics



PRICING STRATEGY

Top 10 Best-Value Products: Deep Discounts Drive Traffic

Exceptional Discounts

Select products offer up to **70%+ discounts**, creating powerful value perception for customers.

Acquisition Tool

Perfect for **deal-of-the-day** promotions and app banners to drive new customer acquisition.

Margin Risk

Monitor carefully to prevent margin erosion from excessive discounting.

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Code JupyterLab Python 3 (ipykernel)

Q1 - Find the top 10 best value products based on discount Percentage.

[19]:
```
df[["name","Category","mrp","discountPercent"]].sort_values(by = "discountPercent", ascending = False).head(10)
```

[19]:

	name	Category	mrp	discountPercent
2619	Dukes Waffy Strawberry Wafers	Biscuits	45.0	0.51
2608	Dukes Waffy Chocolate Wafers	Biscuits	45.0	0.51
2615	Dukes Waffy Orange Wafers	Biscuits	45.0	0.51
1191	RRO Cheddar Block Cheese	Dairy, Bread & Batter	295.0	0.50
1195	RRO Fresh Ricotta	Dairy, Bread & Batter	275.0	0.50
1197	RRO Mozzarella Pizza Cheese	Dairy, Bread & Batter	275.0	0.50
1643	Moi Soi Black Bean Sauce - Dip Spread Stir Fr...	Packaged Food	280.0	0.50
1200	RRO Burrata Cheese	Dairy, Bread & Batter	250.0	0.50
1326	RRO Mozzarella Pizza Cheese	Beverages	275.0	0.50

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Premium Products Out of Stock: Revenue at Risk

The High-Value Gap

Several premium products (MRP >₹500) are currently unavailable, representing a critical business challenge.

1

Strong Demand Signal

Stock-outs indicate robust customer appetite for premium offerings.

2

Forecasting Failure

Poor demand prediction leads to missed opportunities.

3

Revenue Leakage

Every day without stock means lost high-margin sales and dissatisfied customers.

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Code JupyterLab Python 3 (ipykernel)

Q2 - What are the products with high MRP but out of stock?

[26]:
```
df.loc[(df["outOfStock"] == True) & (df["mrp"] > 300)].sort_values(by = "mrp", ascending = False)

#using loc method
```

[26]:

	Category	name	mrp	discountPercent	availableQuantity	discountedSellingPrice	weightInGms	outOfStock	quantity
580	Cooking Essentials	Patanjali Cow's Ghee	565.0	0.00	0	565.0	1000	True	1
1094	Munchies	Patanjali Cow's Ghee	565.0	0.00	0	565.0	1000	True	1
3096	Personal Care	MamyPoko Pants Standard Diapers, Extra Large (...	399.0	0.07	0	369.0	26	True	26
3440	Paan Corner	MamyPoko Pants Standard Diapers, Extra Large (...	399.0	0.07	0	369.0	26	True	26
553	Cooking Essentials	Aashirvaad Atta With Mutigrains	315.0	0.08	0	287.0	5000	True	5
1067	Munchies	Aashirvaad Atta With Mutigrains	315.0	0.08	0	287.0	5000	True	5
573	Cooking Essentials	Everest Kashmiri Lal Chilli Powder	310.0	0.10	0	279.0	500	True	500
1087	Munchies	Everest Kashmiri Lal Chilli Powder	310.0	0.10	0	279.0	500	True	500

Premium products require better demand forecasting and faster replenishment to capture their full revenue potential.

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REVENUE ANALYSIS

Revenue by Category: The Uneven Playing Field

Revenue distribution across categories reveals clear winners and opportunities for strategic focus. Understanding this landscape helps prioritise inventory investments and delivery optimisation.

For a rapid delivery model like Zepto's, understanding granular revenue distribution is not merely a financial exercise; it's fundamental to operational efficiency and customer satisfaction. The 10-minute delivery promise hinges on precise inventory management, optimal dark store layouts, and streamlined picking processes. Misallocating resources to low-volume, low-revenue categories can lead to stock-outs in popular items, increased operational costs due to inefficient space utilization, and ultimately, a failure to meet the core value proposition of speed and reliability. Therefore, a data-driven approach to category performance directly impacts our ability to scale profitably and maintain competitive advantage.

Dominant Categories: Driving Core Business

Categories like fresh produce (milk, eggs), staples (bread, rice, sugar), beverages (soft drinks, water), and packaged snacks consistently drive the highest revenue. These are daily essentials or impulse purchases with high frequency and predictable demand.

For Zepto's 10-minute delivery, these categories are critical. Any stock-out directly impacts customer loyalty and our brand promise. They require deep inventory levels, ensuring constant availability, often with multiple SKUs for popular items. Prioritizing these categories means dedicated, easily accessible slots within dark stores and optimized picking paths to facilitate rapid fulfillment.

Marginal Contributors: Opportunities for Reassessment

This group includes items like specialty imported goods, niche organic products, non-essential household cleaning supplies, and highly seasonal items. While they contribute to product breadth, their individual revenue impact is significantly lower.

Strategic options for these categories include dynamic pricing to stimulate demand, promotional bundling with dominant items to increase perceived value, or even category reassessment for slow-moving, low-margin products. We must explore potential growth opportunities through targeted marketing, expanding into new niche markets, or optimizing supplier agreements to improve margins without compromising the broader product catalog.

Strategic Focus: Optimizing Resource Allocation

Resource allocation must be directly proportional to revenue contribution and operational criticality. This means assigning prime dark store real estate to dominant categories, developing dedicated picking and packing sequences, and aligning logistics resources (delivery riders, dispatch systems) to ensure their fastest possible delivery.

There's a critical trade-off between catalog breadth (offering a wide variety) and inventory depth (having sufficient stock of popular items). For Zepto, depth in dominant categories is paramount for delivery SLAs, while breadth in marginal categories can be managed more leanly. This strategic focus ensures maximum operational efficiency, minimizes waste, and directly impacts our ability to consistently achieve the 10-minute delivery promise.

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Q3 - Calculate estimate revenue for each catagory. ¶

[38]:
```
df.groupby("Category").apply(lambda x : (x["availableQuantity"]*x["discountedSellingPrice"]).sum()).sort_values(ascending = True)

#to calculate, revenue (internal calculation), lambda function was used, we could add new column as revenue and then do groupby too....
```

[38]:
```
Category
Fruits & Vegetables       10846.0
Meats, Fish & Eggs        20693.0
Biscuits                  25007.6
Beverages                 55051.0
Dairy, Bread & Batter     55051.0
Health & Hygiene          64180.0
Home & Cleaning          122661.0
Chocolates & Candies     224385.0
Ice Cream & Desserts     224385.0
Packaged Food            224385.0
Paan Corner              270849.0
Personal Care            270849.0
Cooking Essentials       337369.0
Munchies                 337369.0
dtype: float64
```

Further analysis reveals a significant revenue concentration within a few top categories, often illustrating the Pareto principle where approximately 20% of our categories generate 80% of our revenue. This insight is crucial for Zepto, as it dictates where we should invest the majority of our operational focus and capital.

- **Revenue Concentration:** A deeper dive into sales data confirms that a handful of categories are disproportionately responsible for overall revenue. This concentration is a strength, as it simplifies inventory management for core products, but also a vulnerability if these key categories face supply chain disruptions or increased competition.
- **Pareto Principle Application**: Applying the 80/20 rule allows us to identify the "vital few" categories that underpin our financial success. For these, we must ensure uncompromised stock levels and delivery efficiency. The "trivial many" categories, while contributing to customer choice, require a different, more cost-efficient management approach.
- **Category-Specific Strategies**: Recommendations include aggressive forecasting and safety stock for dominant categories, agile sourcing and potential dynamic pricing for marginal contributors, and a continuous review process for low-performing categories to either optimize them or phase them out. This tiered approach optimizes inventory holding costs, reduces waste, and enhances the overall efficiency of our rapid delivery network.

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Premium Products with Low Discounts: Margin Protectors

Q4 - Find all products where MRP is above 500 and discount is less than 10%.

```
[41]: df.loc[(df["mrp"]>500) & (df["discountPercent"]< 10)].sort_values(by = 'mrp', ascending = False)
```

[41]:

	Category	name	mrp	discountPercent	availableQuantity	discountedSellingPrice	weightInGms	outOfStock	quantity
517	Cooking Essentials	Borges Extra Light Olive Oil Bottle	2600.0	0.46	6	1399.0	2000	False	2
1031	Munchies	Borges Extra Light Olive Oil Bottle	2600.0	0.46	6	1399.0	2000	False	2
3016	Personal Care	Pampers Pants - Large	1549.0	0.45	3	849.0	82	False	82
3360	Paan Corner	Pampers Pants - Large	1549.0	0.45	3	849.0	82	False	82
363	Cooking Essentials	Praakritik Natural Desi Gir Cow A2 Ghee	1450.0	0.10	6	1305.0	540	False	540
...	...	...	...	...	...	...	...	...	...
624	Munchies	Amul Ghee (Tin)	520.0	0.02	6	509.0	1000	False	1
110	Cooking Essentials	Amul Ghee (Tin)	520.0	0.02	6	509.0	1000	False	1
2584	Meats, Fish & Eggs	Fresh Mutton Mince - Kheema (Halal)	515.0	0.03	1	495.0	500	False	500
621	Munchies	Amul Pure Ghee (Pouch)	505.0	0.01	2	499.0	1000	False	1
107	Cooking Essentials	Amul Pure Ghee (Pouch)	505.0	0.01	2	499.0	1000	False	1

124 rows × 9 columns

The Value of Premium Positioning

Products priced above ₹500 with less than 10% discount represent a distinct strategy: **quality over volume**.

01

Brand Value

They leverage strong brand equity.

02

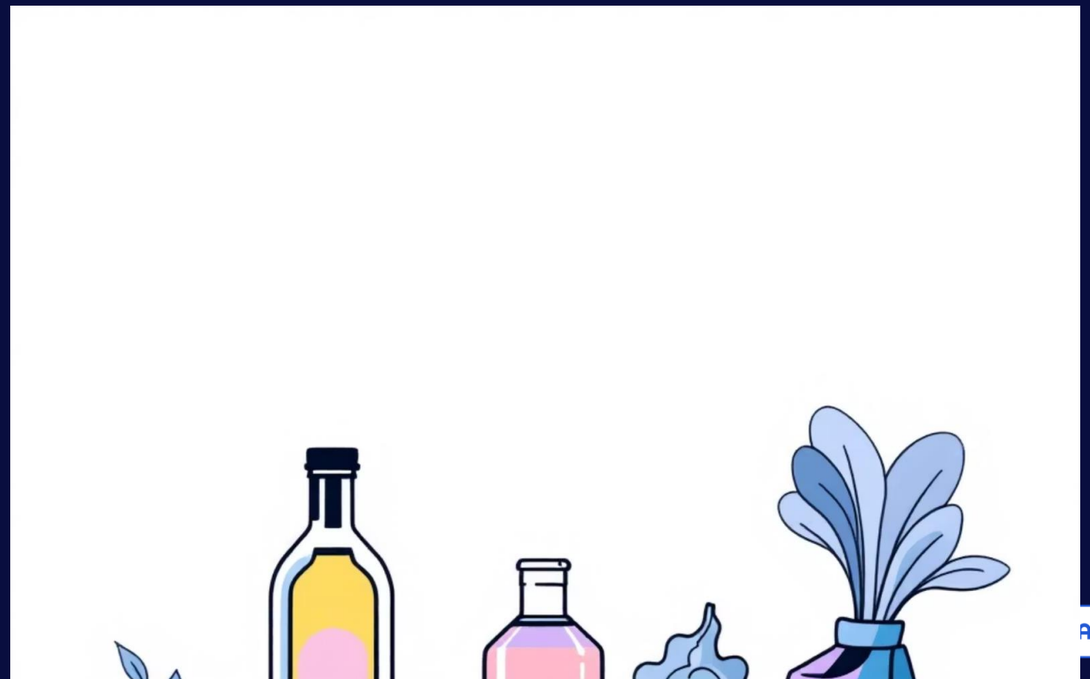
Product Uniqueness

Unique features justify premium pricing.

03

Margin Protection

Profitability is maintained with selective promotions.



Categories with Highest Average Discounts: Competitive Pressure

Certain categories consistently show higher average discounts, reflecting intense market competition and price-sensitive customer behaviour. These categories serve as traffic drivers but require careful margin management.

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Code JupyterLab Python 3 (ipykernel)

Q-5 Identify the top 5 categories offering the highest average discount percentage.

[44]:
```
df.groupby("Category")["discountPercent"].mean().sort_values(ascending = False).head()
```

[44]:
```
Category
Fruits & Vegetables     0.154624
Meats, Fish & Eggs      0.110317
Chocolates & Candies    0.083247
Ice Cream & Desserts    0.083247
Packaged Food           0.083247
Name: discountPercent, dtype: float64
```

1 **Competitive Analysis**

Monitor competitor pricing in high-discount categories to remain competitive without over-discounting.

2 **Supplier Negotiation**

Work with suppliers to secure better terms that protect margins whilst offering customer value.

3 **Traffic Conversion**

Use these categories to drive traffic, then cross-sell higher-margin products.

VALUE METRICS

Best Value by Price per Gram: Bulk Packs Win

The Bulk Advantage

Analysis of products over 100g reveals a clear pattern: larger pack sizes deliver significantly lower price per gram, offering customers genuine cost efficiency.

Identify Best Value

Price-per-gram analysis highlights truly cost-efficient products for budget-conscious shoppers.

Increase Cart Size

Promote bulk packs with "best value" badges to encourage larger purchases.

Boost Satisfaction

Customers appreciate transparent value, driving loyalty and repeat orders.

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Code JupyterLab Python 3 (ipykernel)

Q-6 Find the Price per gram for products above 100 gm and sort by best value.

[50]:
```
df['price_per_gram'] = df["weightInGms"]/df["mrp"]

df.head()
```

[50]:

	Category	name	mrp	discountPercent	availableQuantity	discountedSellingPrice	weightInGms	outOfStock	quantity	price_per_gram
0	Fruits & Vegetables	Onion	25.0	0.16	3	21.0	1000	False	1	40.000000
1	Fruits & Vegetables	Tomato Hybrid	42.0	0.16	3	35.0	1000	False	1	23.809524
2	Fruits & Vegetables	Tender Coconut	51.0	0.15	3	43.0	58	False	1	1.137255
3	Fruits & Vegetables	Coriander Leaves	20.0	0.15	3	17.0	100	False	100	5.000000
4	Fruits & Vegetables	Ladies Finger	14.0	0.14	3	12.0	250	False	250	17.857143

[53]:
```
df.loc[df['weightInGms']>100].sort_values(by = 'price_per_gram', ascending = True)
```

[53]:

	Category	name	mrp	discountPercent	availableQuantity	discountedSellingPrice	weightInGms	outOfStock	quantity	price_per_gram
2909	Personal Care	L'Oreal Paris Excellence Creme Hair Color, 4.2...	620.0	0.00	6	620.0	172	False	172	0.277419
2850	Personal Care	L'Oreal Paris Excellence Creme Hair Color, 1 B...	620.0	0.00	1	620.0	172	False	172	0.277419
3314	Paan Corner	L'Oreal Paris Excellence Creme Hair Color, 4 N...	620.0	0.00	1	620.0	172	False	172	0.277419
2857	Personal Care	L'Oreal Paris Excellence Creme Hair Color, 3 D...	620.0	0.00	6	620.0	172	False	172	0.277419
3253	Paan Corner	L'Oreal Paris Excellence Creme Hair Color, 4.2...	620.0	0.00	6	620.0	172	False	172	0.277419
...	...	...	...	...	...	...	...	...	...	...
609	Munchies	Tata Salt	24.0	0.00	6	24.0	1000	False	1	41.666667
20	Fruits & Vegetables	Onion	68.0	0.16	3	57.0	3000	False	3	44.117647
1064	Munchies	Aashirvaad Iodised Salt	22.0	0.13	0	19.0	1000	True	1	45.454545
550	Cooking Essentials	Aashirvaad Iodised Salt	22.0	0.13	0	19.0	1000	True	1	45.454545
3644	Health & Hygiene	Vicks Cough Drops Menthol	20.0	0.00	6	20.0	1160	False	20	58.000000

2660 rows × 10 columns

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Product Segmentation & Inventory Weight Insights

Products naturally segment into three distinct groups based on quantity: Low (convenience), Medium (regular household needs), and Bulk (value and family purchases). Understanding these segments alongside inventory weight patterns reveals operational opportunities.

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Code JupyterLab Python 3 (ipykernel)

Q-7 Group the products into catagories like low, medium, Bulk.

[55]:
```
#creating NewCatagory Column

labels = ['low','medium','high']

df['NewCatagory'] = pd.qcut(df['quantity'], q = 3, labels = labels)

df.head()
```

[55]:

	Category	name	mrp	discountPercent	availableQuantity	discountedSellingPrice	weightInGms	outOfStock	quantity	price_per_gram	NewCatagory
0	Fruits & Vegetables	Onion	25.0	0.16	3	21.0	1000	False	1	40.000000	low
1	Fruits & Vegetables	Tomato Hybrid	42.0	0.16	3	35.0	1000	False	1	23.809524	low
2	Fruits & Vegetables	Tender Coconut	51.0	0.15	3	43.0	58	False	1	1.137255	low
3	Fruits & Vegetables	Coriander Leaves	20.0	0.15	3	17.0	100	False	100	5.000000	medium
4	Fruits & Vegetables	Ladies Finger	14.0	0.14	3	12.0	250	False	250	17.857143	medium

Q-8 What is total inventory weight per catagory?

[58]:
```
df.groupby("Category").apply(lambda x : (x["weightInGms"]*x["availableQuantity"]).sum())
```

[58]:
```
Category
Beverages                143735
Biscuits                  84431
Chocolates & Candies     490797
Cooking Essentials      1404654
Dairy, Bread & Batter    143735
Fruits & Vegetables       91794
Health & Hygiene         142904
Home & Cleaning          373161
Ice Cream & Desserts     490797
Meats, Fish & Eggs        48016
Munchies                1404654
Paan Corner              348187
Packaged Food            490797
Personal Care            348187
dtype: int64
```

Inventory Weight Impact

Categories with disproportionately large inventory weight affect storage costs and last-mile delivery efficiency.

Segmentation Strategy

- **Low quantity:** Impulse buys and convenience purchases
- **Medium quantity:** Regular household staples
- **Bulk quantity:** Value seekers and family households

Personalisation

Use segmentation for tailored product recommendations

Subscriptions

Design repeat-order strategies by segment

Warehouse Optimisation

Allocate space based on weight and turnover

KEY INSIGHTS

Final Takeaways for Zepto's Growth Strategy

1

Category-Specific Discounting

Tailor discount strategies by category rather than applying uniform policies—protect margins where brand strength allows.

2

Premium Product Availability

Ensure high-MRP products remain in stock through improved demand forecasting and faster replenishment cycles.

3

Bulk Pack Promotion

Leverage bulk products' superior value proposition to increase average order value and customer satisfaction.

4

Inventory Optimisation

Balance inventory weight with turnover rates to control storage costs whilst maintaining delivery speed.

Data-driven decisions fuel Zepto's rapid expansion and customer delight

By leveraging inventory analytics, Zepto can continue to deliver on its 10-minute promise whilst building a sustainable, profitable business model that delights customers and captures market share in India's competitive quick commerce landscape.

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